

```

atlas atlas-fpmd08-shard-8 [primary] test> db.book_metadata.find(
  {
    avgRating: { $gt: 4 }
  }
)
[
  {
    _id: ObjectId('643305a1b7910a6634027b03'),
    title: 'The Silent Algorithm',
    author: 'A. Benson',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'A. Khan',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1801',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1842',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('643307ba07910a6634027b04'),
    title: 'The Silent Algorithm',
    author: 'A. Benson',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'A. Khan',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1801',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  }
]

```

```

atlas atlas-fpmd08-shard-8 [primary] test> db.book_metadata.insertMany(
  [
    {
      title: 'Waves of Change',
      author: 'B. Thompson',
      genre: 'Technology',
      publishedYear: 2021,
      formats: [ 'Hardcover', 'Audiobook' ],
      specs: {
        pages: 210,
        fileSize_MB: 2.1,
        drm: false
      },
      avgRating: 4.2,
      reviews: [
        { member_id: 'M1801', rating: 4, comment: 'Great Digital read, clear layout.', date: ISODate('2021-06-04') },
        { member_id: 'M1842', rating: 5, comment: 'Best programming book this year.', date: ISODate('2021-06-01') }
      ]
    },
    {
      title: 'Wonders of Technology',
      author: 'C. Brown',
      genre: 'Science',
      publishedYear: 2019,
      formats: [ 'Hardcover', 'Digital' ],
      specs: {
        pages: 410,
        binding: 'Leather'
      },
      avgRating: 3.8,
      reviews: [
        { member_id: 'M1801', rating: 3, comment: 'Nice book but strong smell.', date: ISODate('2019-12-12') },
        { member_id: 'M1842', rating: 5, comment: 'Beautifully written piece.', date: ISODate('2019-01-12') }
      ]
    },
    {
      title: 'Quantum Horizons',
      author: 'D. Green',
      genre: 'Science',
      publishedYear: 2023,
      formats: [ 'Digital', 'Audiobook' ],
      specs: {
        fileSize_MB: 1.8,
        drm: true,
        narrator: 'G. Lee',
        runtime_min: 140
      },
      avgRating: 4.9,
      reviews: [
        { member_id: 'M1801', rating: 5, comment: 'The Digital format made navigation so easy.', date: ISODate('2023-06-02') },
        { member_id: 'M1842', rating: 5, comment: 'Mind-blowing see well narrated.', date: ISODate('2023-06-14') }
      ]
    }
  ]
)
SyntaxError: Unexpected token, expected "," (38:17)
   18 |     title: 'Wonders of Technology',
   19 |     author: 'C. Brown',
>  38 |     genre: 'Science',
      |           ^
   39 |     publishedYear: 2019,
   40 |     formats: [ 'Hardcover', 'Audiobook' ],
   41 |     specs: {
   42 |       pages: 410,
   43 |       binding: 'Leather'
   44 |     },
   45 |     avgRating: 3.8,
   46 |     reviews: [
   47 |       { member_id: 'M1801', rating: 3, comment: 'Nice book but strong smell.', date: ISODate('2019-12-12') },
   48 |       { member_id: 'M1842', rating: 5, comment: 'Beautifully written piece.', date: ISODate('2019-01-12') }
   49 |     ]
   50 |   },
   51 |   {
   52 |     title: 'Quantum Horizons',
   53 |     author: 'D. Green',
   54 |     genre: 'Science',
   55 |     publishedYear: 2023,
   56 |     formats: [ 'Digital', 'Audiobook' ],
   57 |     specs: {
   58 |       fileSize_MB: 1.8,
   59 |       drm: true,
   60 |       narrator: 'G. Lee',
   61 |       runtime_min: 140
   62 |     },
   63 |     avgRating: 4.9,
   64 |     reviews: [
   65 |       { member_id: 'M1801', rating: 5, comment: 'The Digital format made navigation so easy.', date: ISODate('2023-06-02') },
   66 |       { member_id: 'M1842', rating: 5, comment: 'Mind-blowing see well narrated.', date: ISODate('2023-06-14') }
   67 |     ]
   68 |   }
   69 | ]

```

```
Atlas atlas-fpmd0-shard-0 [primary] test> db.book_metadata.find({
  avgRating: { $gt: 4 }
});
[
  {
    _id: ObjectId('6a83d6a1b7910b06f54427b3'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a83d7bab7910b06f54427b4'),
```

```
Atlas atlas-fpmdu0-shard-0 [primary] test> db.book_metadata.find({
|   formats: { $in: ["Digital", "Audiobook"] }
| });
[
  {
    _id: ObjectId('6a83d6a1b7910b06f54427b3'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
]
```

```

Atlas atlas-fpmd0-shard-0 [primary] test> db.book_metadata.find({
|   avgRating: { $gt: 4 },
|   formats: { $in: ["Digital"] }
| });
[
  {
    _id: ObjectId('6a83d6a1b7910b06f54427b3'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  }
]

```

```

Atlas atlas-fpmd0-shard-0 [primary] test> db.book_metadata.find(
|   {
|     formats: {
|       $regex: "Digital",
|       $options: "i"
|     }
|   },
|   {
|     title: 1,
|     formats: 1
|   }
| );
[
  {
    _id: ObjectId('6a83d6a1b7910b06f54427b3'),
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  {
    _id: ObjectId('6a83d7bab7910b06f54427b4'),
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  {
    _id: ObjectId('6a83d7c2b7910b06f54427b5'),
    title: 'Ocean of Code',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  }
]

```

```

Atlas atlas-fpmd00-shard-0 [primary] test> db.book_metadata.find(
  {
    "reviews.comment": {
      $regex: "Digital",
      $options: "i"
    }
  },
  {
    title: 1,
    "reviews.$": 1
  }
);
[
  {
    _id: ObjectId('6a83d6a1b7910b06f54427b3'),
    title: 'The Silent Algorithm',
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a83d7bab7910b06f54427b4'),
    title: 'The Silent Algorithm',
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  }
]

```

```
]
Atlas atlas-fpmdu0-shard-0 [primary] test> db.book_metadata.aggregate([
|   {
|     $match: {
|       publishedYear: { $gt: 2020 }
|     }
|   },
|   {
|     $unwind: "$reviews"
|   },
|   {
|     $group: {
|       _id: "$title",
|       avgReviewRating: {
|         $avg: "$reviews.rating"
|       },
|       totalReviews: {
|         $sum: 1
|       },
|       author: {
|         $first: "$author"
|       }
|     }
|   },
|   {
|     $sort: {
|       avgReviewRating: -1
|     }
|   },
| ]
```

```

Atlas atlas-fpmdu0-shard-0 [primary] test> db.book_metadata.createIndex({
|   "reviews.comment": "text"
| });
reviews.comment_text
Atlas atlas-fpmdu0-shard-0 [primary] test> db.book_metadata.find(
|   $text: {
|     $search: "Digital"
|   }
| });
[
|   {
|     _id: ObjectId('6a83d7bab7910b06f54427b4'),
|     title: 'The Silent Algorithm',
|     author: 'R. Menon',
|     genre: 'Technology',
|     publishedYear: 2022,
|     formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
|     specs: {
|       pages: 340,
|       fileSize_MB: 4.2,
|       drm: true,
|       narrator: 'K. Rao',
|       runtime_min: 610
|     },
|     avgRating: 4.6,
|     reviews: [
|       {
|         member_id: 'M1001',
|         rating: 5,
|         comment: 'Excellent deep dive into digital systems design.',
|         date: ISODate('2023-01-15T00:00:00.000Z')
|       },
|       {
|         member_id: 'M1002',
|         comment: 'Best programming book this year.',
|         date: ISODate('2022-06-01T00:00:00.000Z')
|       }
|     ]
|   }
| ]

```

```

Atlas atlas-fpmdu0-shard-0 [primary] test> db.book_metadata.find(
|   {
|     $text: {
|       $search: "Digital"
|     }
|   },
|   {
|     title: 1,
|     score: {
|       $meta: "textScore"
|     }
|   }
| ).sort({
|   score: {
|     $meta: "textScore"
|   }
| });
[
|   {
|     id: ObjectId('6a83d7bab7910b06f54427b4')
|   }
| ]

```

```
stage: 'FETCH',
nReturned: 8,
executionTimeMillisEstimate: 0,
works: 9,
advanced: 8,
needTime: 0,
needYield: 0,
saveState: 0,
restoreState: 0,
isEOF: 1,
docsExamined: 8,
alreadyHasObj: 0,
inputStage: {
  stage: 'IXSCAN',
  nReturned: 8,
  executionTimeMillisEstimate: 0,
  works: 9,
  advanced: 8,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  keyPattern: { _fts: 'text', _ftsx: 1 },
  indexName: 'reviews.comment_text',
  isMultiKey: true,
  isUnique: false,
  isSparse: false,
  isPartial: false,
  indexVersion: 2,
  direction: 'backward',
  indexBounds: {},
  keysExamined: 8,
  seeks: 1,
  dupsTested: 8,
  dupsDropped: 0
}
```

```

command: {
  find: 'book_metadata',
  filter: { '$text': { '$search': 'Digital' } },
  '$db': 'test'
},
serverInfo: {
  host: 'ac-jypisep-shard-00-02.njclshi.mongodb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787025460, i: 1 }),
  signature: {
    hash: Binary.createFromBase64('zg+foD6Xj6NSQbZ52zf15nEh7Vw=', 0),
    keyId: Long('7616375491519315970')
  }
},
operationTime: Timestamp({ t: 1787025460, i: 1 })
}
atlas atlas-fpmdu0-shard-0 [primary] test> db.book_metadata.find({
  "reviews.comment": {
    $regex: "Digital",
  }
})

```